
Linear Algebra and Its Applications (MAT2041)

Course Syllabus and Brief Introduction

Instructor:

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数据科学学院

School of Data Science

Introduction

- Basic Information and Course Syllabus
- Basic Introduction

Course Information (L03)

Instructor: Ruoyu Sun

(you may call me Prof. Sun, or Professor Sun, by university convention)

Venue: TxC 203

Office Hour: TBD. (TBD—may change after Feb)

Email: sunruoyu@cuhk.edu.cn

Important notice & QA: Wechat group
(add during the break)

Homework: Blackboard

Discussion: Piazza (TBD)

Online Platform

Wechat group: for receiving important notice and material
(e.g. Slides, homework)

& asking important logistics question; not Tech questions
(e.g. cannot find homework on BB)

Piazza: for discussion of tech questions (TBD)

These links will be
shared in wechat
group, and pinned
on the top
(会在微信群里分
享并置顶)

Blackboard: submitting homework, grading homework

Myself

- Peking University, math, BS
- Univ of Minnesota, ECE, PhD
- Stanford University, Postdoc
- Facebook AI Research (led by Yann LeCun), visiting scientist
- UIUC, Assistant Professor

- CUHK(SZ), Associate professor. **Starting Fall 2022.**
- **Research Areas:** machine learning, optimization and communication networks, including:
large language models, deep learning theory, adaptive algorithms, learning to optimize, graph neural networks, resource allocation in networks, ...

- I taught an early version of linear algebra in 2017 in CUHK-SZ summer, with Prof. Tom Luo.
- I taught MAT2041 in Fall 2022 and Fall 2023, Spring 2025.

2026 TA and USTF

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No Tutorial for the first week

Textbooks

Teaching Method: slides + writing

Textbook: [GS] Gilbert Strang, *Introduction to Linear Algebra*, 5th or 6th edition, Wellesley-Cambridge Press

Recommended books: [Boyd] Steven Boyd, Lieven Vandenberghe, *Introduction to Applied Linear Algebra*
[Leon] Steven J. Leon, *Linear Algebra with Applications*, 9th Edition
[Lay] David C. Lay, et al., *Linear Algebra and its Applications*, 5th edition

Note: The textbook is not the same as the contents.

We use material from the textbook and other recommended books.

Gilbert Strang & Art of Teaching Linear Algebra

Gilbert Strang: MIT professor;

- **Iconic Textbook [GS]**

- highlighting geometry and insight

- filled with real-world examples from multiple fields

- **Core Philosophy:** Focuses on "**Understanding, not just computation.**"

- **Lasting Impact:** shapes how linear algebra is taught today.

Other books

- **Steven Boyd**: former Dean of Engineering, Stanford University,
- Not satisfied with Stanford's linear algebra teaching, so wrote his own textbook, with many modern examples
- [Boyd]: very special kind of book; quite different from all other books

[Leon] Steven J. Leon, *Linear Algebra with Applications*, 9th Edition

[Lay] David C. Lay, et al., *Linear Algebra and its Applications*, 5th edition

Classical math type books.

Contain lots of formal definitions, lemmas, theorems

Relation of Course and Books

- The outline and around 60-80% of course contents are from [GS].
- A bit from [Boyd]; but followed the application spirit of [Boyd]
- Some math definitions and lemmas follow [Leon] and [Lay].
- **Remark:** [GS] lacks formal definitions & theorems;
we have a bit more rigorous results than [GS].

Grading Scheme

Attendance:	10%	(checked after 3 rd week, end of add/drop period) --using an app
Assignments and Quiz:	25%	(In total approx. 7 assignments, 4-5 quizzes)
Mid-term Exam:	30%	
Final-term Exam:	35%	Final exam covers all material of the semester.

Attendance

- There will be an app for signing-in
- We will share the QR code
- When you arrive at the classroom, you scan the code and click “sign-in”; the system will record
- Checked after 3rd week, end of add/drop period

Quiz and Assignment

- **Assignment policy:** Drop 1 worst assignment score.
- **Quiz:** roughly 15-20 mins; in-class
(likely during tutorial session; wait for announcement);
closed-book quiz;
answer questions on the papers (not online).
randomly drawn from the set of review questions
- If you cannot attend the quiz, need to ask for permission from instructor;
TA/USTF will arrange an make-up quiz

Other Logistics

- Homework 1: will be due at least next week (probably Friday or so).
- **Dropping one homework policy:** worst score can be dropped
- **Submit via BB**

Course Outline (tentative)

Logistics + Motivation (data science) + Vectors (norms, inner products)	Week 1
Linear Systems I: row/column forms, elimination, Gauss–Jordan, ERO	Week 2
Matrix Algebra: multiplication/properties, transpose, block/partitioned products	Week 3
Linear Systems & Matrices + Vector Space I: elementary matrices, inverse/GE, permutation, LU/PLU, REF/RREF, complexity, 2D solution picture, $Ax=0$, col/null spaces	Week 4
Linear System Solution Theory + Vector Space II: $Ax=b$ solvability, particular/general solutions; independence, basis, dimension, rank, four fundamental subspaces	Week 5
Vector Space III + Applications + Review: adjacency matrix, convolution, recommendation systems	Week 6
Orthogonality Basics: orthogonal set, orthogonal complement	Week 7
Four Fundamental Subspaces + Midterm (covers up to Lec 15) + Least Squares (linear regression, projection)	Week 8
Orthogonality + Determinants: orthogonal matrices, orthonormal basis, Gram–Schmidt; determinants intro + properties + geometric interpretation	Week 9–10
Linear Transformations: definitions, properties, similarity, matrix representation	Week 10–11
Eigen-Theory: eigenvalues/eigenvectors, characteristic polynomial, spectral decomposition, diagonalization/matrix powers, spectral theorem, non-diagonalizable; PageRank	Week 11–13
Singular Value Decomposition I–II	Week 13–14
Quadratic Forms: quadratic forms, PD matrices, Cholesky	Week 14

Tip: English Vocabulary and Piazza

- Students can check this vocabulary, and edit yourself
【腾讯文档】 MAT2041 Vocabulary

<https://docs.qq.com/doc/DWHFWbnIWZUhEdUF5>